

## Unidirectional Flushing (UDF) Plan – NAS I Sigonella

(\*) Reference Section C, Annex 0200000 Management and Administration, Spec Item 2.6.2 Dissemination of Information

This plan outlines the work required to perform the “Unidirectional flushing” at NAS I. This results in water flowing toward the hydrant in only one direction resulting in higher flow velocity, more scouring and better cleaning of the pipes with less water use than the “Conventional flushing” This plan will be implemented once per year.

### Safety procedure.

Apply before and during the UDF.

1. Necessary personal protective equipment and the correct tools:
  - a. Personal Protective Equipment (PPE)
    - i. Safety shoes
    - ii. Hard hat
    - iii. Hearing protection
    - iv. Safety glasses
    - v. Protective gloves
    - vi. Reflective vest (CAT 2)
  - b. Site Safety Equipment Required
    - i. Availability of communications (telephone or radio)
2. Employees tasked to complete flushing tasks need be minimum two-person due to the distances between hydrants and valves.
3. At least one crew member should be designated to control traffic while the other crew members are operating the distribution system valves.
4. Traffic signs, cones, flags, and/or vehicles with warning lights should be used as necessary to alert motorists of flushing activities.
5. Brightly colored safety vests shall be worn by all crew members.
6. Keep the public away from flushing activities. **Children are often attracted to such activities and should be kept away.**
7. Open hydrant valves completely to prevent water from discharging through the barrel drain at pressure.
8. Open and close hydrants slowly to prevent the development of dangerous pressure surges and water-hammer.
9. Use well-restrained energy dissipaters designed to prevent the water pouring out of the valve from damaging government and private property.
10. Discharge flushing water directly into a sanitary or storm sewer whenever possible to avoid flooding streets and underground electrical vaults. If street flooding is unavoidable, use signs, flags, and other warning devices to direct traffic away from the area. Notify Security and Fire Department about the UDF situation.

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The desired minimum velocity is 6 feet per second. A single 2-1/2 inch outlet should be sufficient for the 6-inch and 8-inch mains. For larger mains, it may be more advantageous to open additional hydrants and open the 4-1/2 inch nozzle. Utilize pitot gauges with flow diffusers to minimize grounds damage.

| Main Size | Suggested Minimum Flow |
|-----------|------------------------|
| (in)      | (gpm)                  |
| 6         | 550                    |
| 8         | 950                    |
| 10        | 1500                   |
| 12        | 2100                   |
| 14        | 2800                   |

Note: In most cases, customers will not notice any changes to their water delivery during the UDF process, but UEM staff must inform the Fire Department and the sensitive customer (Hospital, school etc.) before to start the UDF.

Loop 1: (\*) (Total of 6 Valves to Be Secured)

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W22A  | 315  | (*)      |
| W24A  | 315  |          |
| W138A | 200  |          |
| W13A  | 200  |          |
| W157A | 200  |          |
| W17A  | 200  |          |

Listed below is the sequence of hydrants to operate. Secure each hydrant before proceeding to the next location. Flow each hydrant until the flowing water is clear. Record the approximate minutes of flushing performed at each location.

| Sequence | Hyd ID | Location |
|----------|--------|----------|
| 1        | FH1A   | (*)      |
| 2        | FH37A  |          |
| 3        | FH40A  |          |
| 4        | FH41A  |          |
| 5        | FH44A  |          |
| 6        | FH29A  |          |
| 7        | FH20A  |          |
| 8        | FH19A  |          |
| 9        | FH28A  |          |
| 10       | FH24A  |          |
| 11       | FH23A  |          |

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Valves to be reopened:

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W138A | 200  | (*)      |
| W13A  | 200  |          |
| W157A | 200  |          |
| W17A  | 200  |          |

Loop 2: (\*) (Total of 1 Valve to Be Secured 2 valves already secured during loop1 )

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W22A  | 315  | (*)      |
| W24A  | 315  |          |
| W152A | 150  |          |

Listed below is the sequence of hydrants to operate. Secure each hydrant before proceeding to the next location. Flow each hydrant until the flowing water clears. Record the approximate minutes of flushing performed at each location.

| Sequence | Hyd ID | Location |
|----------|--------|----------|
| 1        | FH14A  | (*)      |
| 2        | FH46A  |          |
| 3        | FH13A  |          |
| 4        | FH11A  |          |
| 5        | FH2A   |          |
| 6        | FH10A  |          |
| 7        | FH4A   |          |
| 8        | FH6A   |          |
| 9        | FH35A  |          |

Valves to be reopened:

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W22A  | 315  | (*)      |
| W24A  | 315  |          |
| W152A | 150  |          |

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Loop 3: (\*) (Total of 2 Valves to Be Secured)

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W28A  | 315  | (*)      |
| W35A  | 315  |          |

Listed below is the sequence of hydrants to operate. Secure each hydrant before proceeding to the next location. Flow each hydrant until the flowing water clears. Record the approximate minutes of flushing performed at each location.

| Sequence | Hyd ID | Location |
|----------|--------|----------|
| 1        | FH2A   | (*)      |
| 2        | FH35A  |          |

Valves to be reopened:

| Valve | Size | Location |
|-------|------|----------|
|       | (mm) |          |
| W28A  | 315  | (*)      |
| W35A  | 315  |          |

(\*)